



FINAL YEAR PROJECT

SMART MINUTES

AI-powered meeting recorder & summarizer

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BS Information Technology

Rawalpindi Women University

8.3%

Word error
rate

4.2/5

AI summary
quality

1.4s

Caption
latency

3

Role
tiers

What we'll cover



What is Smart Minutes?

An AI-powered desktop application that automates the entire meeting documentation lifecycle — from live recording to structured, distributed minutes.



Live & post-meeting STT

Real-time captions via Vosk; high-accuracy transcripts via OpenAI Whisper



AI summarization

Google Gemini 1.5 Pro generates structured minutes with decisions and topics



Offline + online

Supports both in-person and remote meetings in one platform



Three-tier RBAC

Super Admin, Admin, and User roles for organizational governance



Modern stack

Flutter desktop frontend with a FastAPI / Python backend



Incremental delivery

Built using the Incremental Life Cycle Model across 5 stages



Why meetings need a smarter system



Time-Consuming

Manual note-taking is time-consuming and distracts participants



No Offline Support

No tool offers full offline support for in-person meetings



Weak Access Control

Current solutions lack role-based access and org management



Missed Action Items

Decisions and action items are frequently missed or forgotten



Cloud-Only Tools

Existing tools (Otter.ai, Fireflies.ai, Teams) are cloud-only



Fragmented Workflow

No system unifies task extraction, tracking, and PDF minutes



The Smart Minutes solution



Live Transcription

Real-time + post-meeting transcription
(Vosk + Whisper)



Role-Based Access

Three-tier role-based access control



Semantic Insights

Semantic analysis: tone, topic relevance,
keywords



AI-Generated Minutes

Structured minutes generation via Google
Gemini 1.5 Pro



Hybrid Storage

Hybrid offline/online support with secure
storage



One-Click Reporting

Professional PDF reports with one-click email
sharing



Smart Task Management

Automatic task extraction, assignment &
notifications

Scope



Target users

Corporate, academic, and government organizations



Platforms

Windows 10/11, macOS 12+, Ubuntu 20.04+ (desktop only)



Speaker ID

Based on text context, not audio characteristics



Language

English only Vosk and Whisper configured for English



Accuracy limits

May drop in high-background-noise environments



Internet dependency

AI summarization requires internet (Gemini API)

Literature review



Vosk

Offline, Kaldi-based ASR with multilingual support



OpenAI Whisper

Trained on 680k hours of audio; robust to accents & noise



Google Gemini 1.5 Pro

Long-context NLP for summarization & extraction



Otter.ai

Online only — no role management or formal minutes



Fireflies.ai

Subscription-based — no offline support



Google Meet / MS Teams

Cloud-locked — no PDF minutes generation

Smart Minutes is the only solution offering hybrid offline+online support, RBAC, and full org management

Comparison with existing systems

Feature	Smart Minutes	Otter.ai	Fireflies.ai	Google Meet	MS Teams	Justification
Offline support	Yes	No	No	No	Limited	Smart Minutes uses Vosk + Whisper locally; all others require internet
Online support	Yes	Yes	Yes	Yes	Yes	Smart Minutes added Firebase-based online meeting support
AI summarization	Full	Basic	Moderate	No	Limited	Gemini 1.5 Pro extracts structured decisions, actions, topics
Role-based access	Full	No	No	No	Limited	3-tier RBAC: Super Admin, Admin, User with full permissions
Task management	Full	Moderate	Moderate	No	Limited	Auto task extraction, assignment, tracking, notifications
Meeting minutes	Full	Basic	Moderate	No	Limited	Full formatted minutes with PDF export; others produce transcripts only
Semantic analysis	Full	Basic	Moderate	Basic	Basic	Tone detection, topic relevance, keyword extraction via Gemini

Tools & technologies



Frontend

Flutter 3.24+ with Dart — cross-platform desktop UI



Backend

Python 3.9+, FastAPI, Uvicorn ASGI server



AI / ML

Vosk, OpenAI Whisper, Google Gemini 1.5 Pro API



Data

MySQL 8.0, SQL Alchemy ORM, Firebase Auth, JWT



Real-time

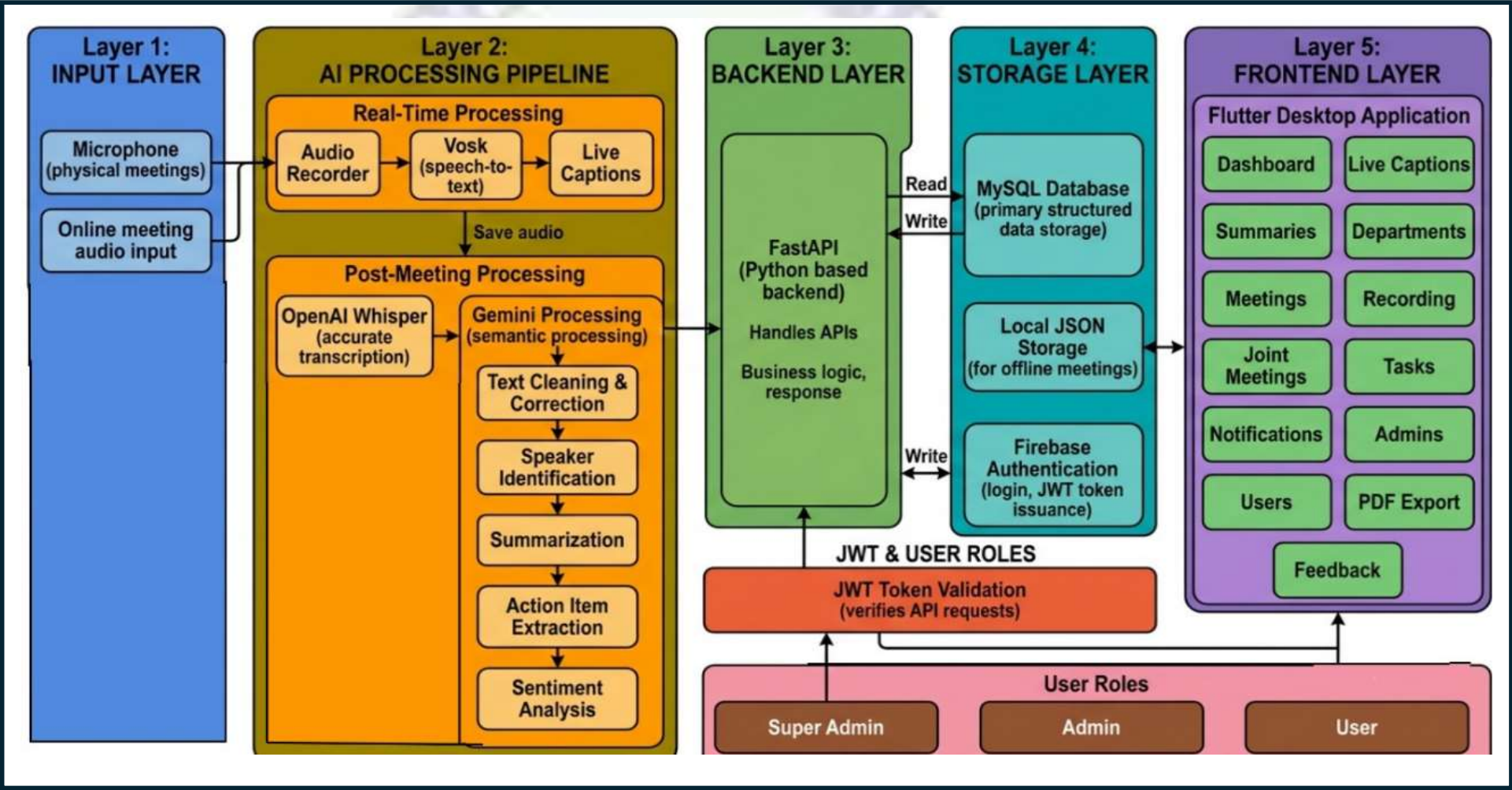
WebSocket & WebRTC for live audio streaming



Processing

FFmpeg for audio processing & WAV merging

System architecture



System modules



Authentication

Login, registration, role assignment, sessions



Meeting management

Create, view, edit, delete, start meetings



Audio recording

Start/stop, WAV merging, disconnect handling



Live transcription

Real-time captions, full transcript export



AI summarization

Generate, edit, approve/reject (Admin only)



Task management

Auto-extract, assign, track, notify



User & dept mgmt

Role assignment, department creation



Reports & analytics

PDF generation, email, dashboard stats

Functional & non-functional requirements

F

Functional

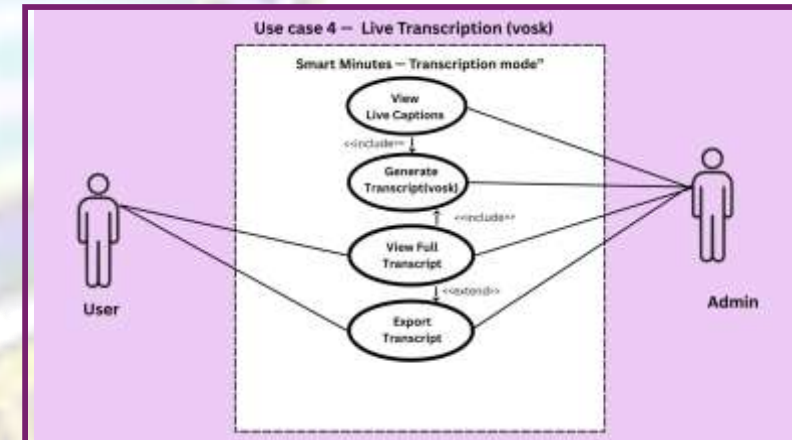
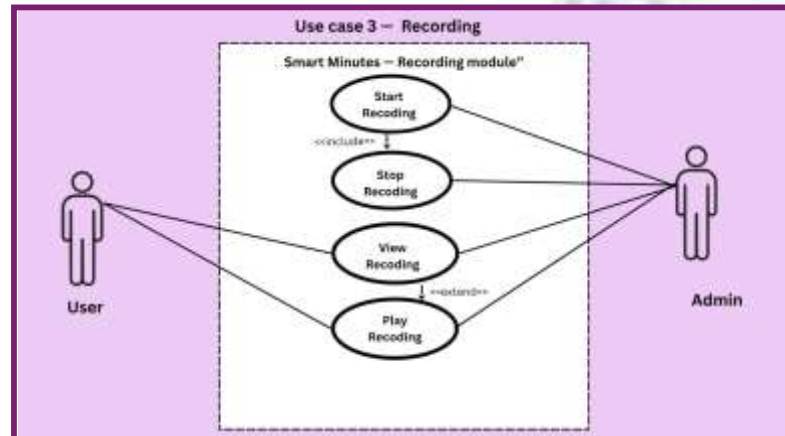
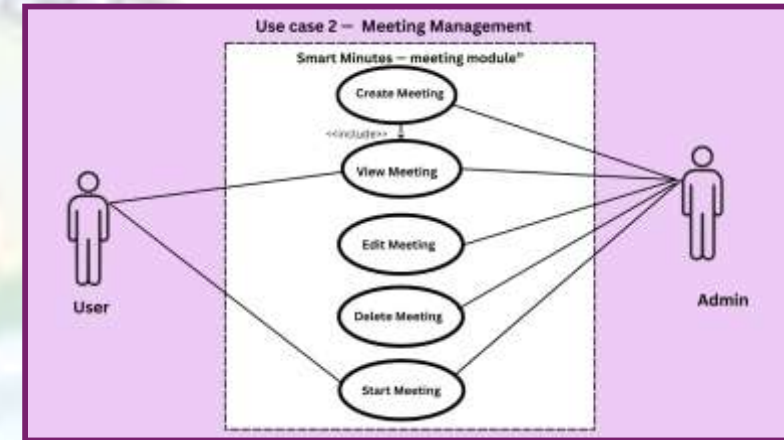
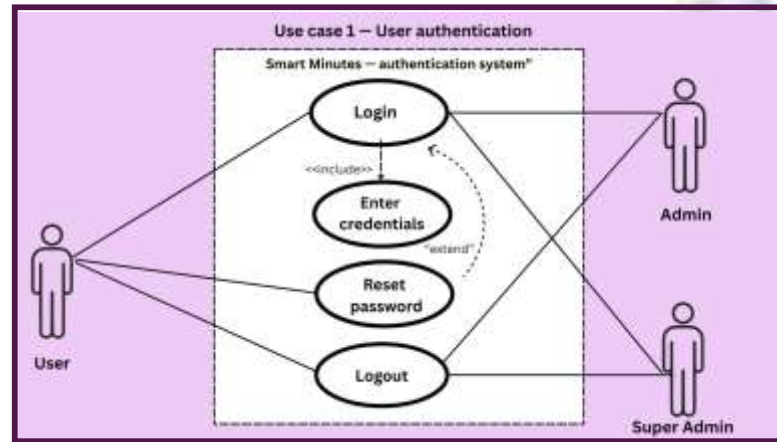
- Schedule, organize & track meetings with agenda
- Role-based secure access (Super Admin, Admin, User)
- Live capture with real-time captions & ASR transcripts
- Auto-generate summaries with decisions & action items
- Admin review, edit, and approval of AI-generated minutes
- Auto task creation, assignment & PDF + email distribution

N

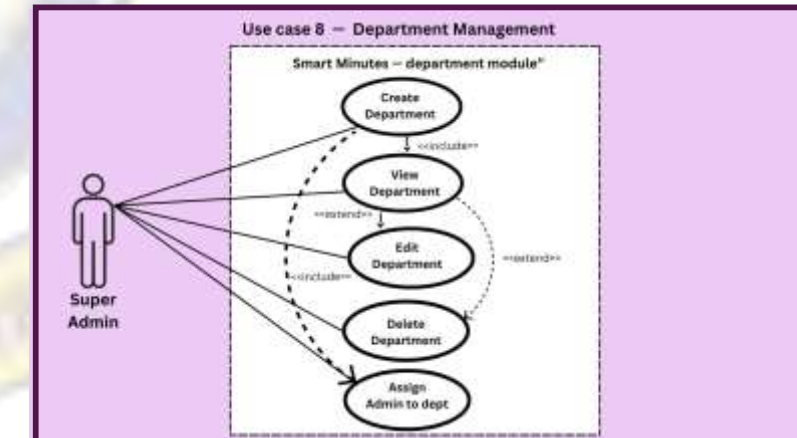
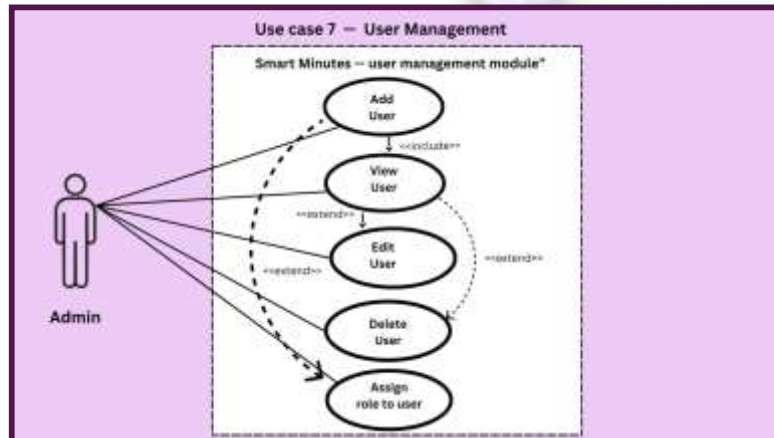
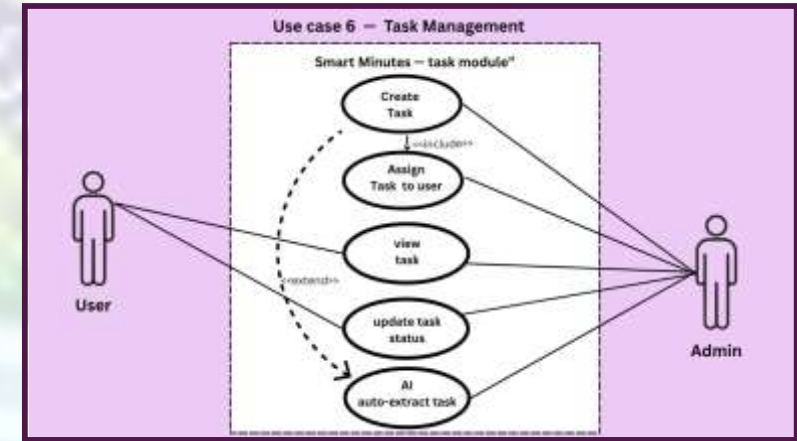
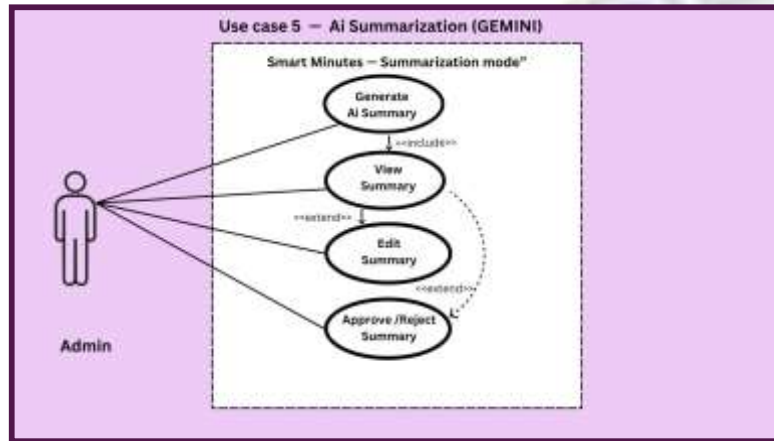
Non-functional

- Word Error Rate under 15%
- Live captions delivered within 2 seconds
- Cross-platform: Windows, macOS, Linux
- Secure: Firebase + JWT + RBAC enforced
- Usable by both technical and non-technical users

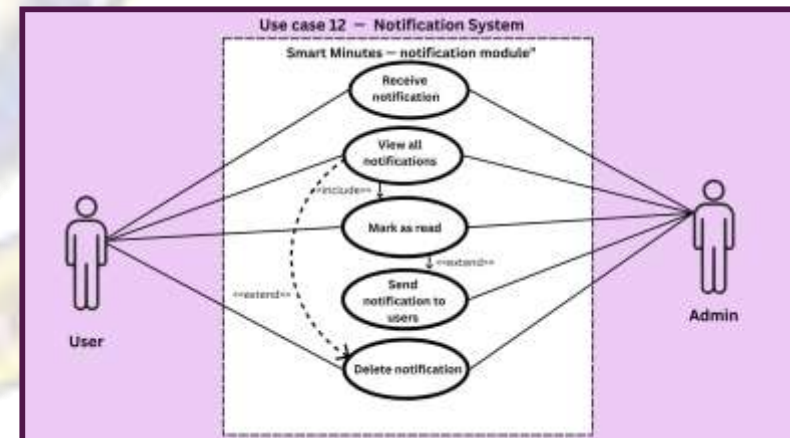
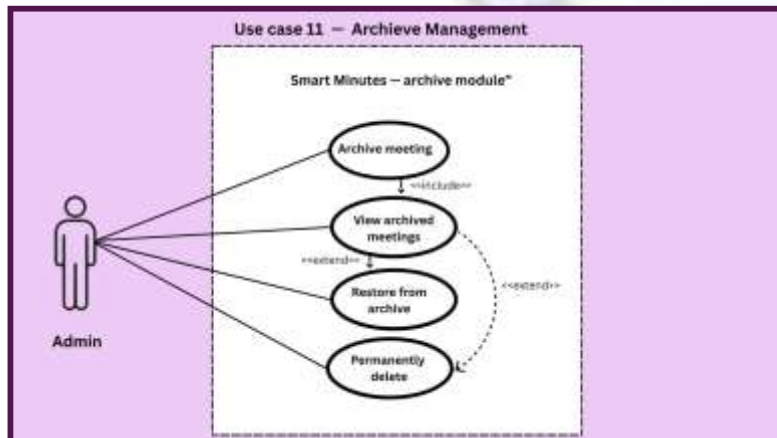
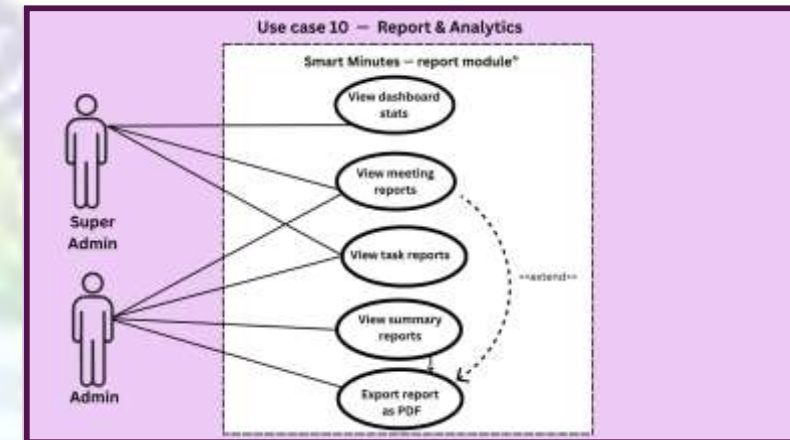
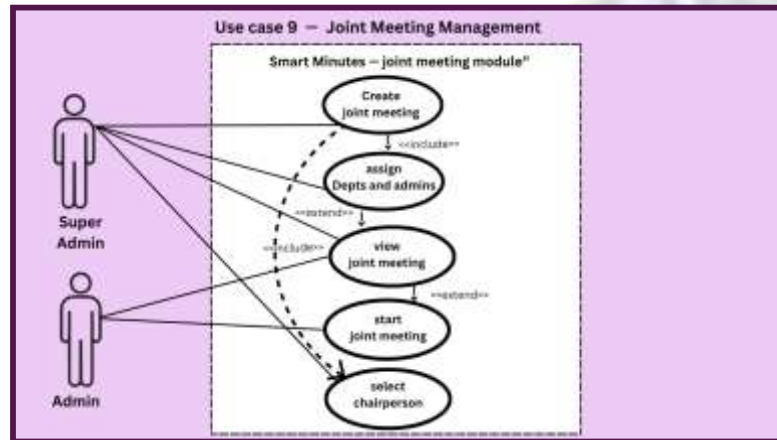
Use case diagrams (1 - 4)



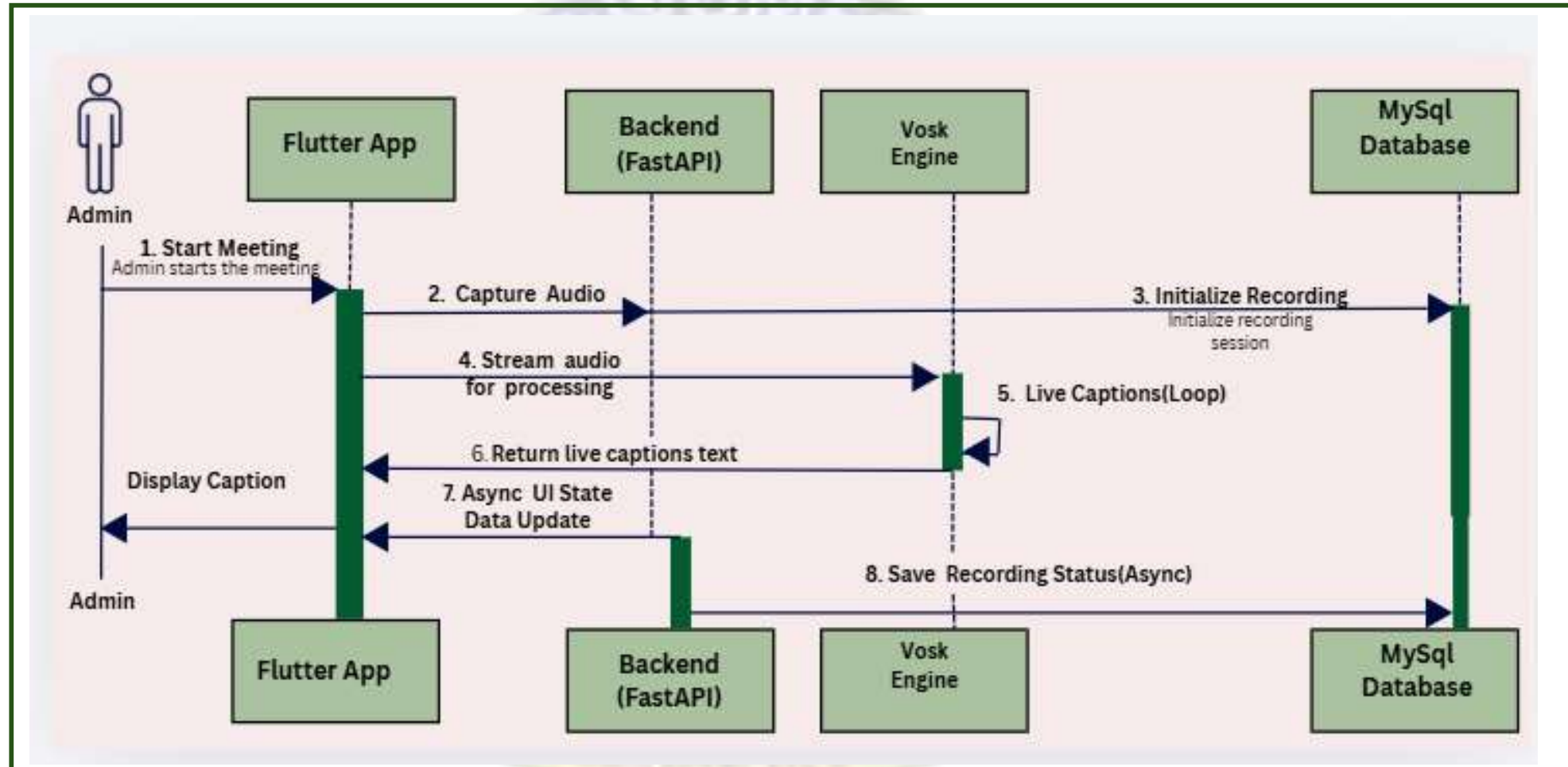
Use case diagrams (5 - 8)



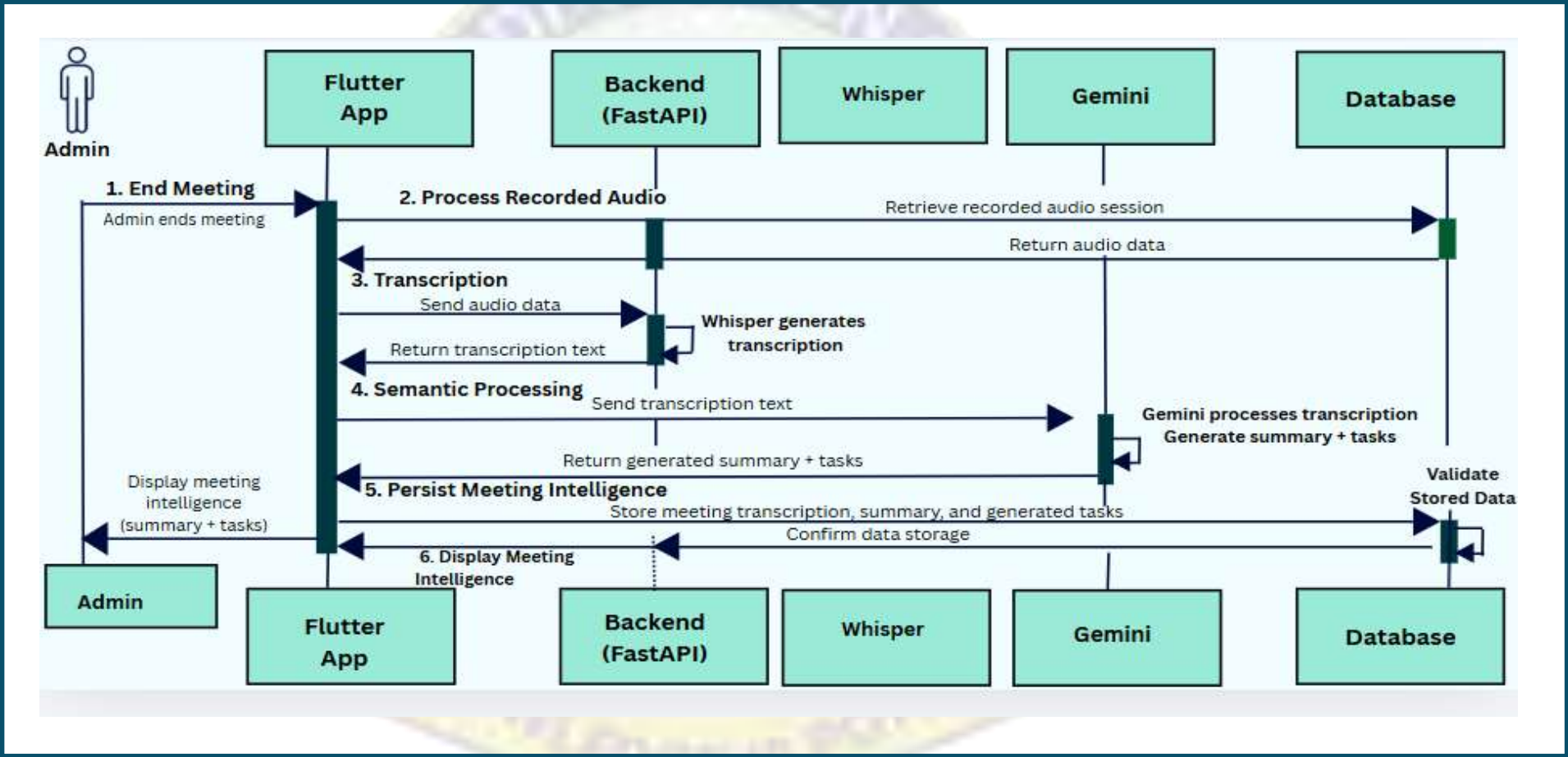
Use case diagrams (9 - 12)



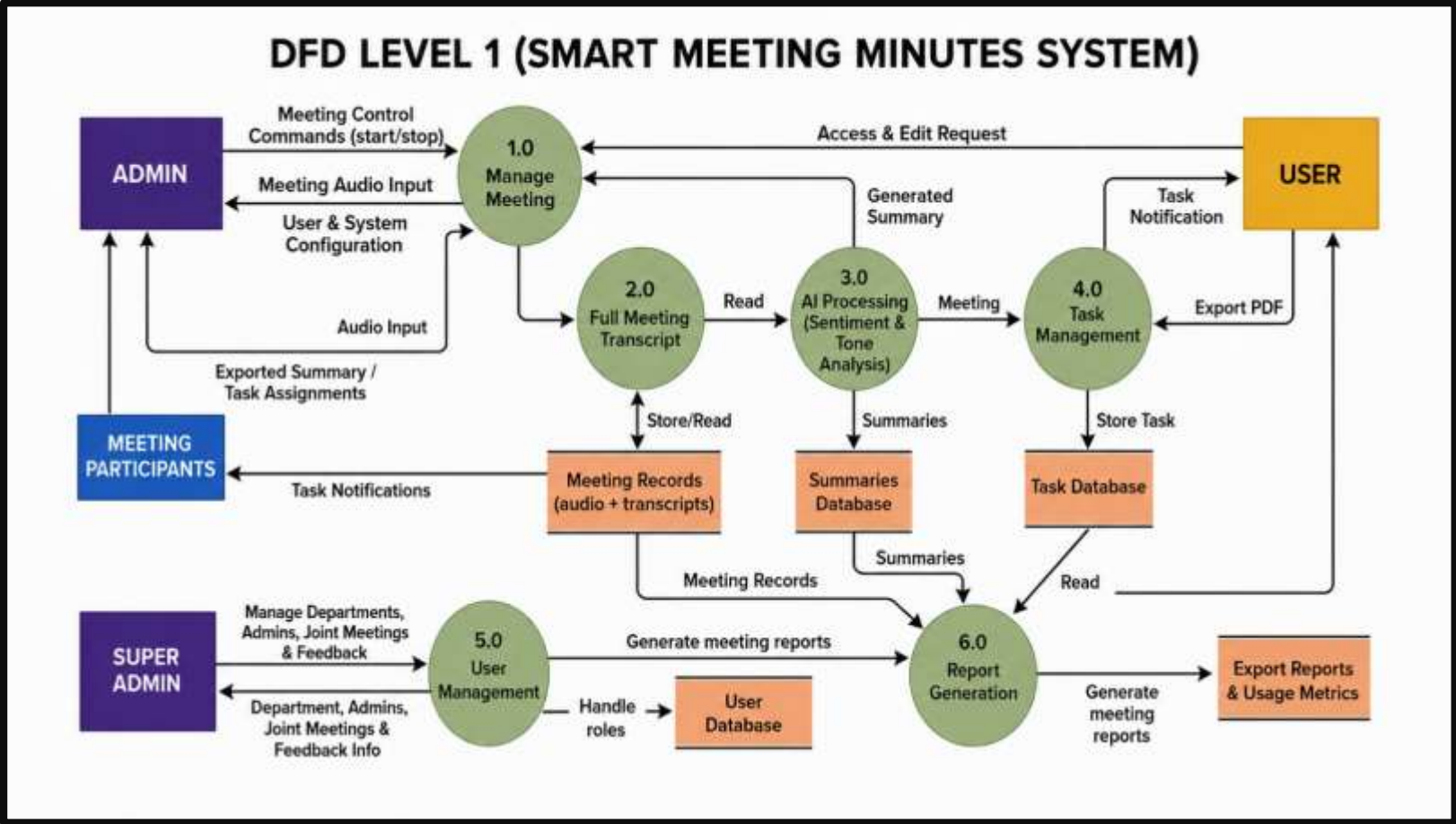
Sequence diagram - live recording



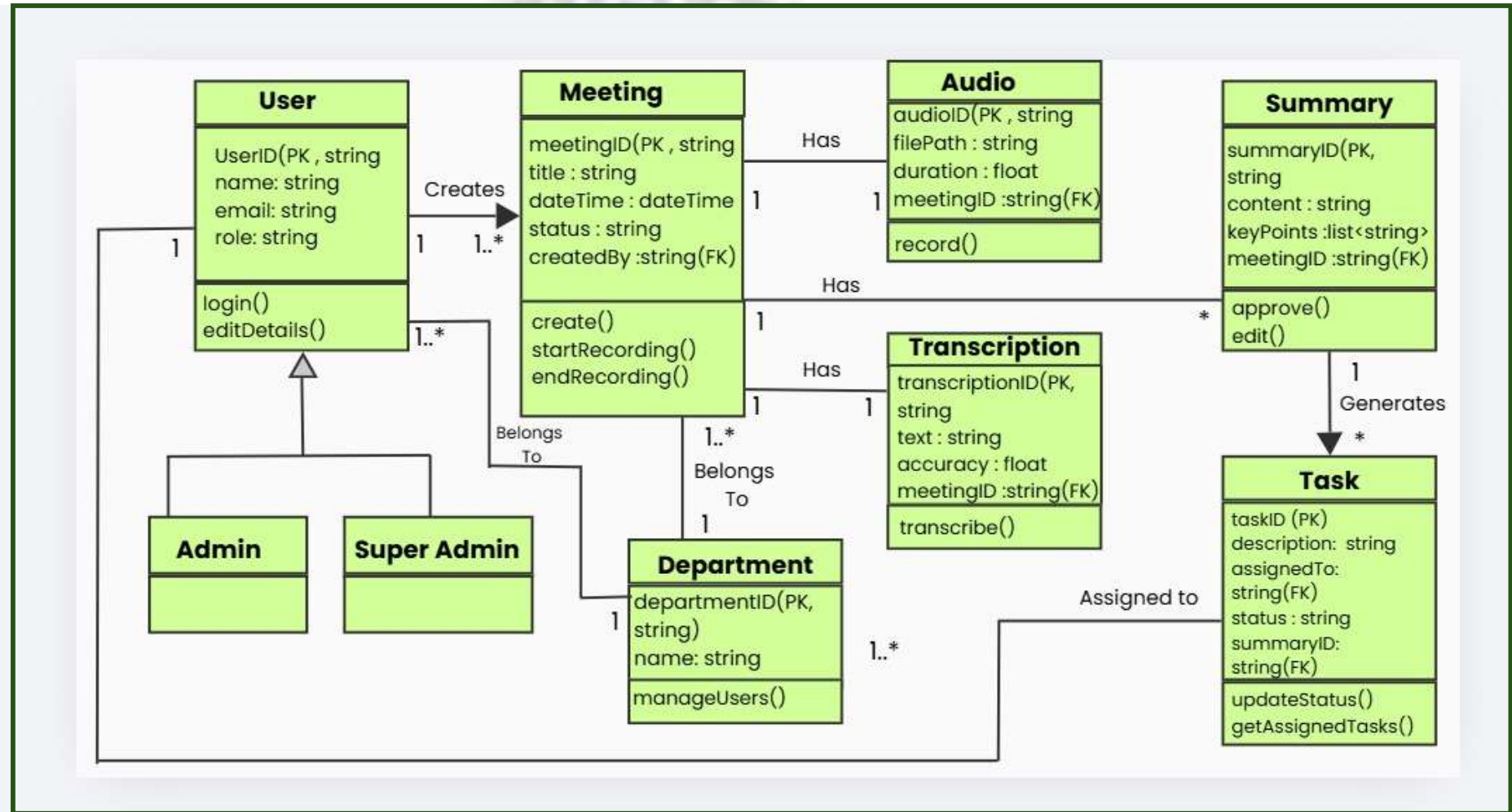
Sequence diagram - AI processing



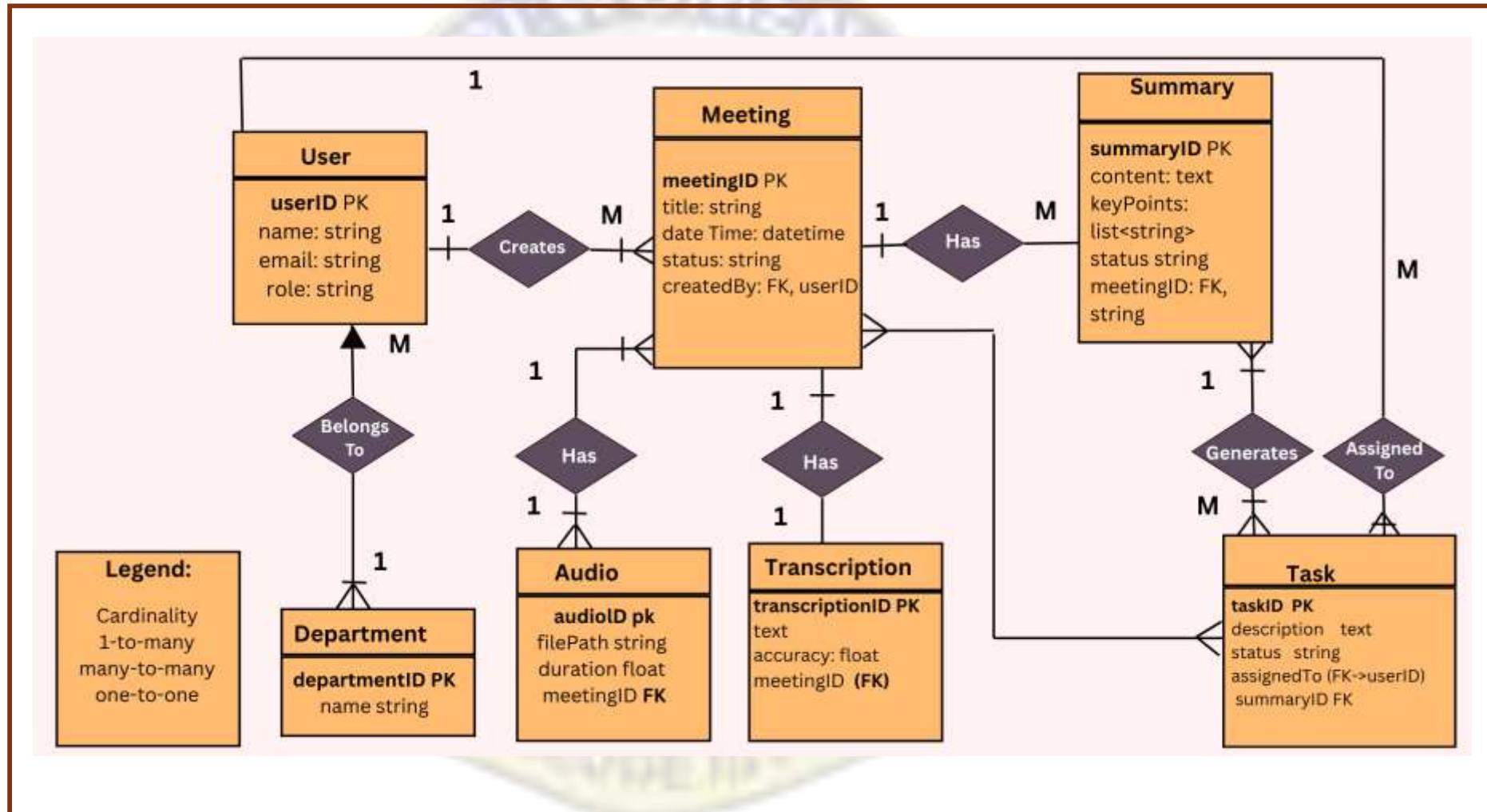
Data flow diagram - level 1



Class diagram



Entity relationship diagram



Survey results - 114 responses

57%

Find auto-minutes extremely
valuable

56.1%

Definitely/probably will use
Smart Minutes

64%

Say it would genuinely help
them

69.3%

Have never used an AI
meeting tool

Key survey insights



Note-taking method

52.6% manually write notes - 23.7% typed during meetings 13.2% audio recorded - 7% no formal method



Time wasted per meeting

41.2% waste 20-30 min - 37.7% waste 10-20 min - 19.3% waste 30+ minutes on manual documentation



Adoption intent

56.1% definitely/probably will use Smart Minutes - Only 1.8% said they definitely won't



Would it help?

64% said YES it would genuinely help their org - 28.9% said maybe - very high positive signal



AI tool experience

69.3% have never used an AI meeting tool - huge untapped market - Key issues: internet, accuracy, language



Top requested features

Multi-language (Urdu) support - Calendar integration-Speaker identification - Reminder system

User interface

Built with Flutter 3.24+ - a single cross-platform desktop codebase for Windows, macOS, and Linux.



Login screen

Firestore-integrated, role-based
secure access



Dashboard

Role-specific layout for each
user type



Meeting screen

Record button, live captions,
status indicator



Summary review

Admin can view, edit, approve
or reject AI summaries



Task manager

View tasks, update status,
get notifications



Reports

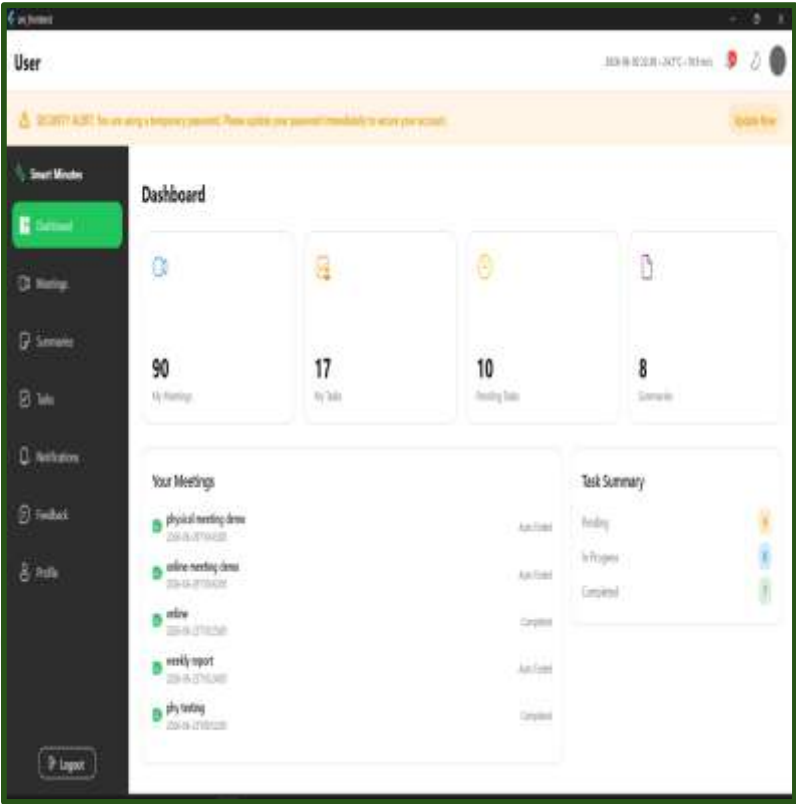
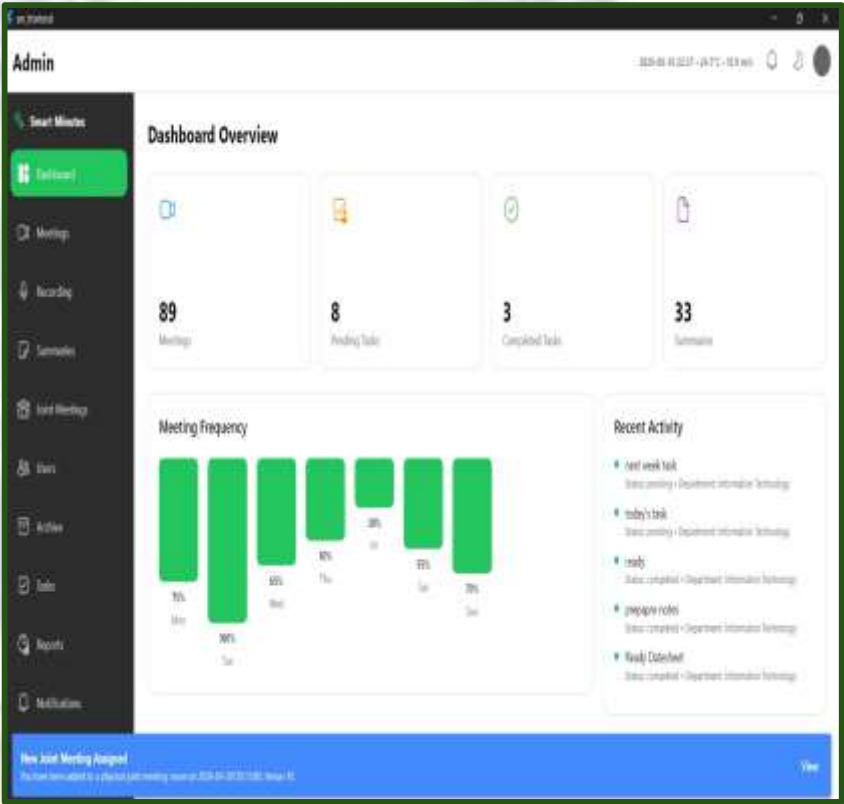
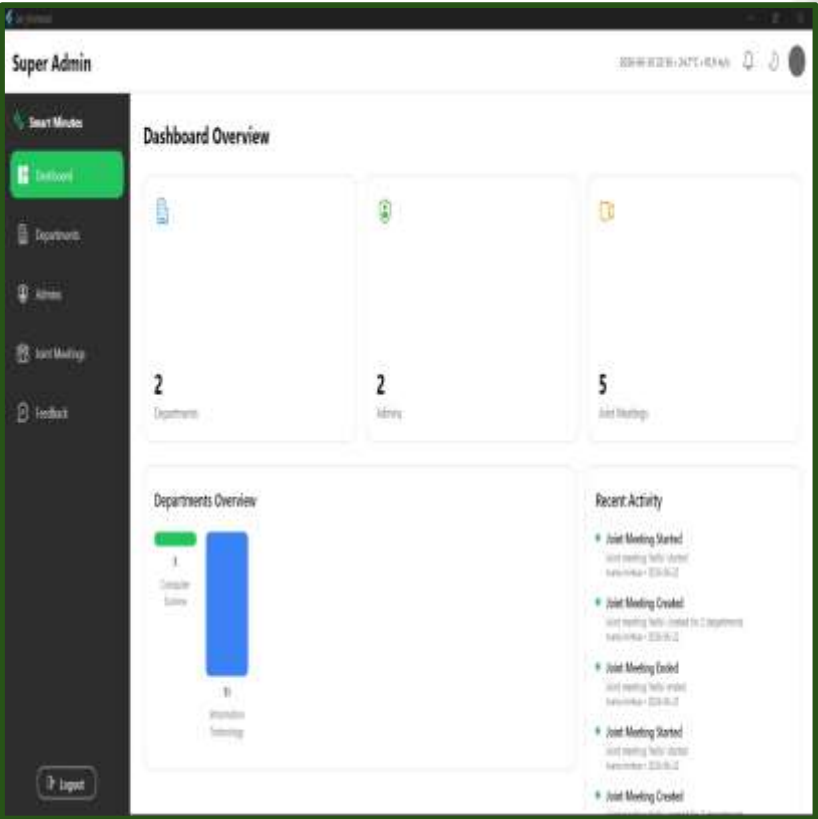
PDF preview with one-click
email distribution

Login screen

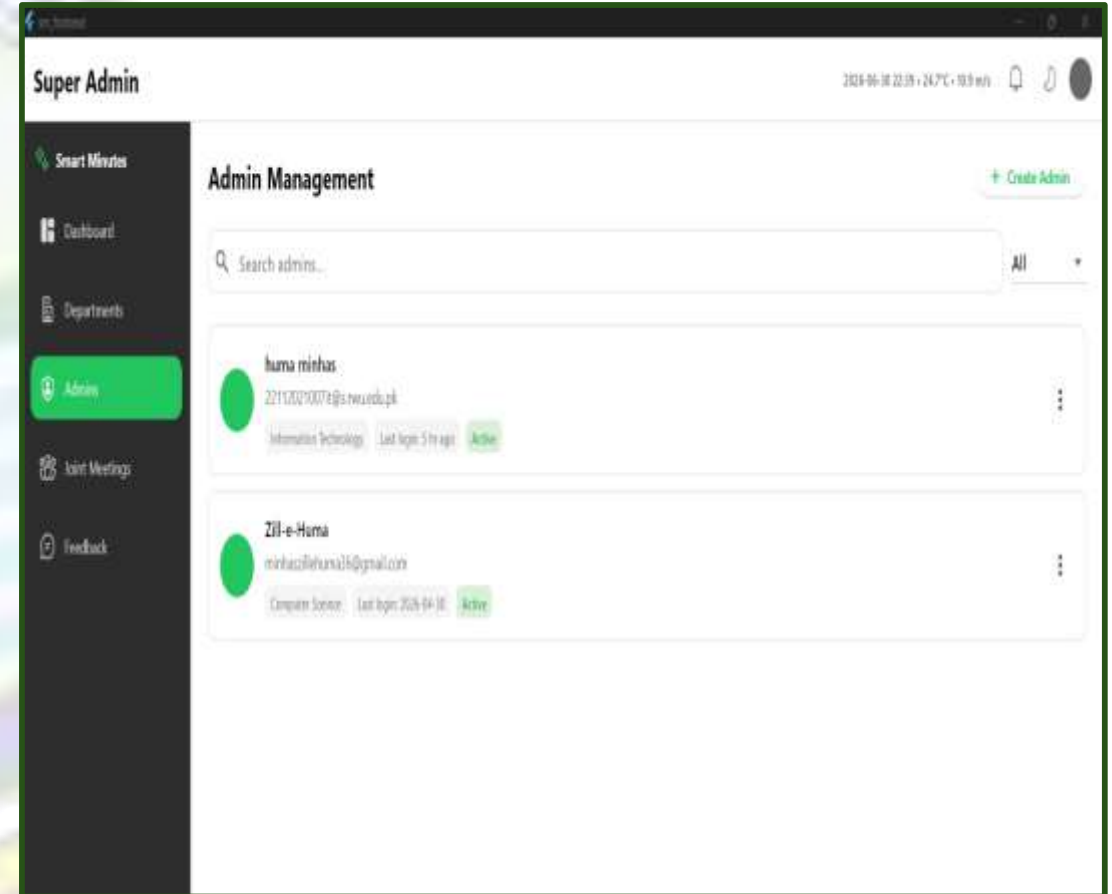
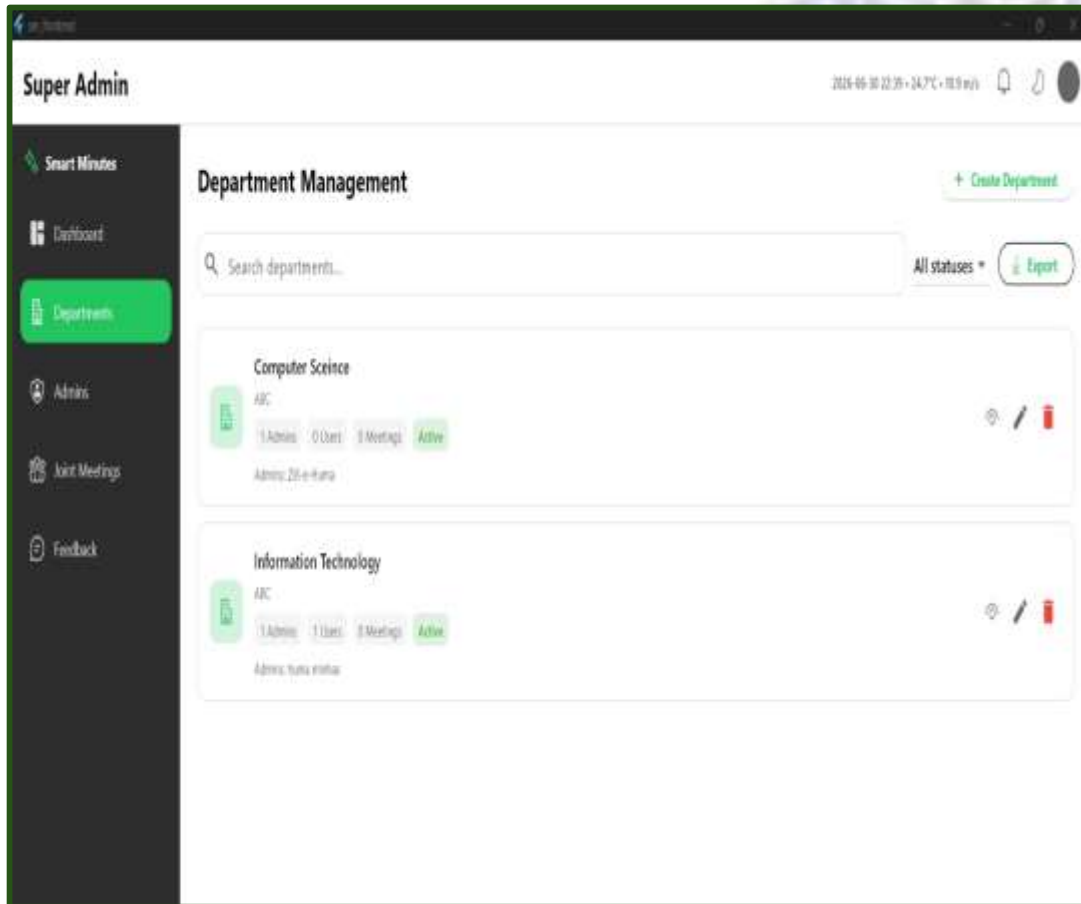
The screenshot shows a web application window titled 'sm_frontend'. The login screen has a 'Welcome Back' heading and a sub-header 'Sign in to continue to Smart Minutes'. The main login area contains an 'Email' input field with the value '22112021007it@s.rwu.edu.pk', a 'Password' input field with a toggle icon, a checked 'Remember Me' checkbox, a 'Forgot Password?' link, and a 'Login' button. Below the login fields is a 'Demo Credentials' section with the text 'Use credentials sent by email after account provisioning.' To the right is a 'Quick Login' section with four entries, each featuring a green circular profile icon, an email address, a role, and 'x'/'>' icons.

Email	Role
humaminhas14314@gmail.com	SUPER_ADMIN
22112021007it@s.rwu.edu.pk	ADMIN
aansahfayyaz13388@gmail.com	USER
minhaszillehuma36@gmail.com	ADMIN

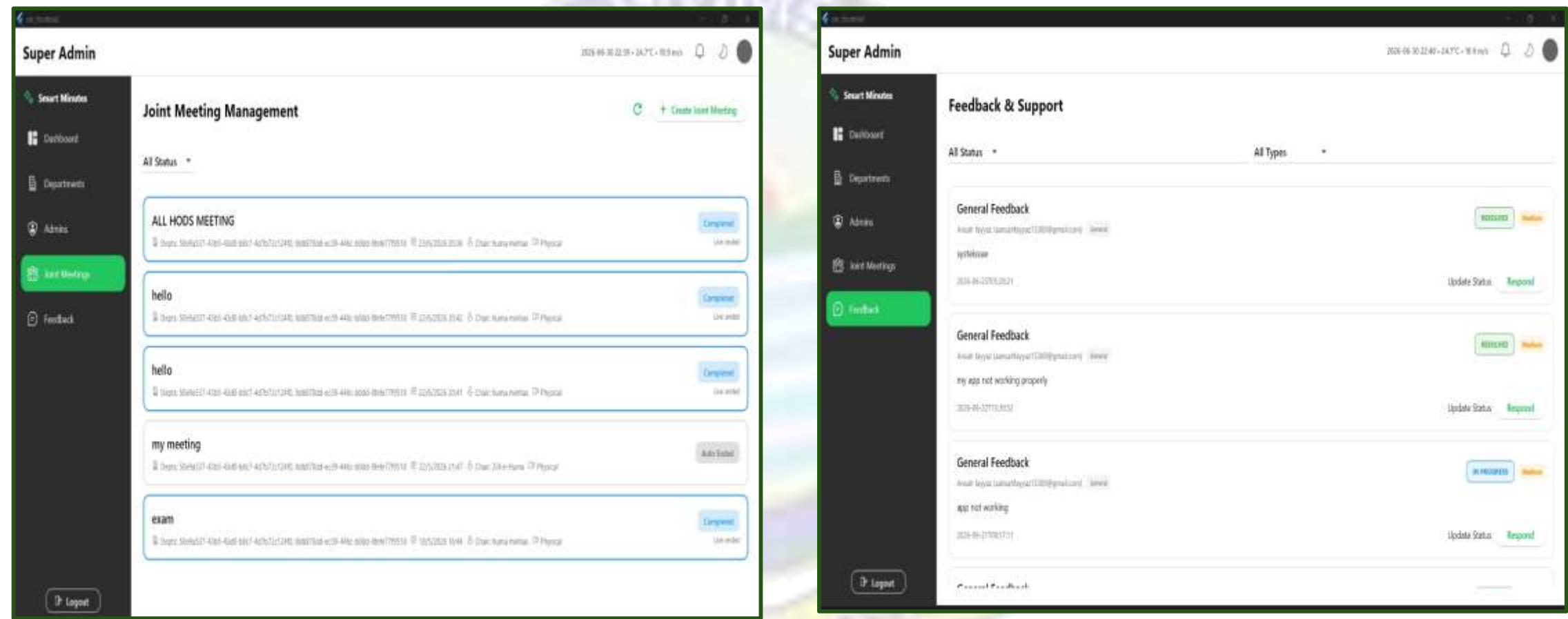
Role-based dashboards



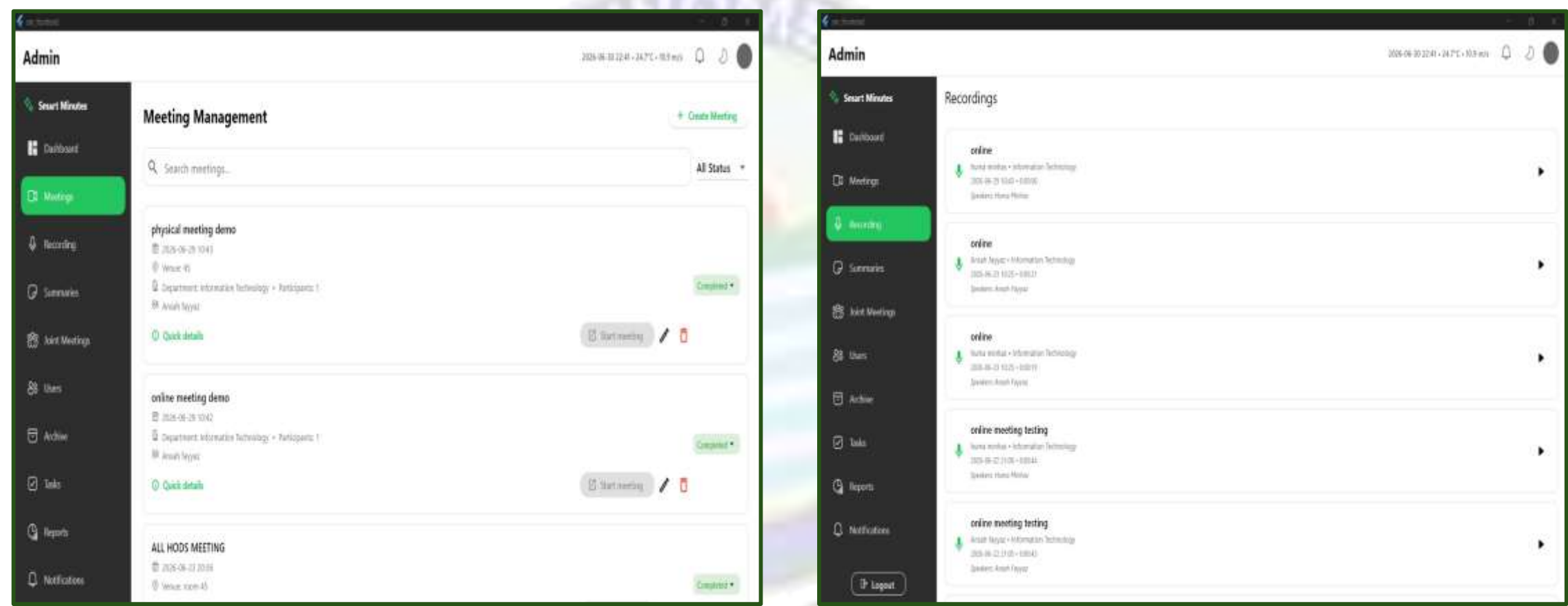
Department & admin management



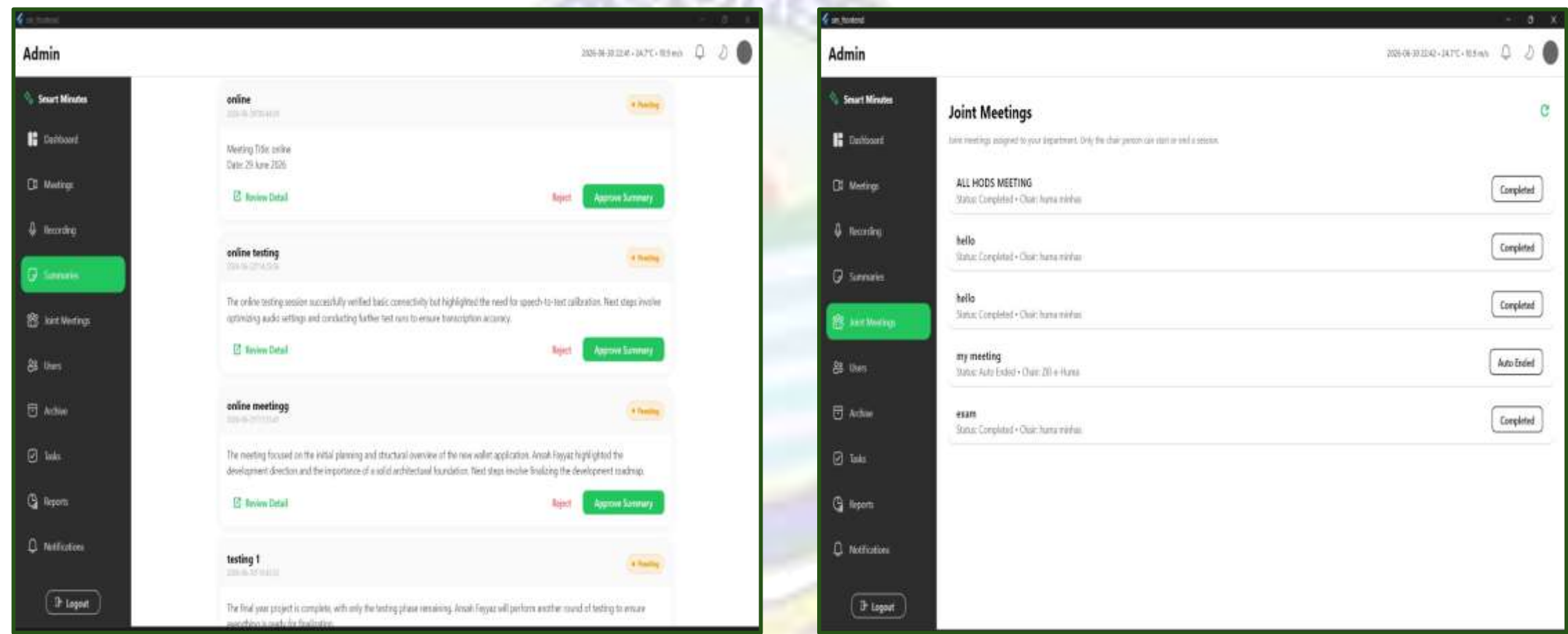
Joint meetings & feedback



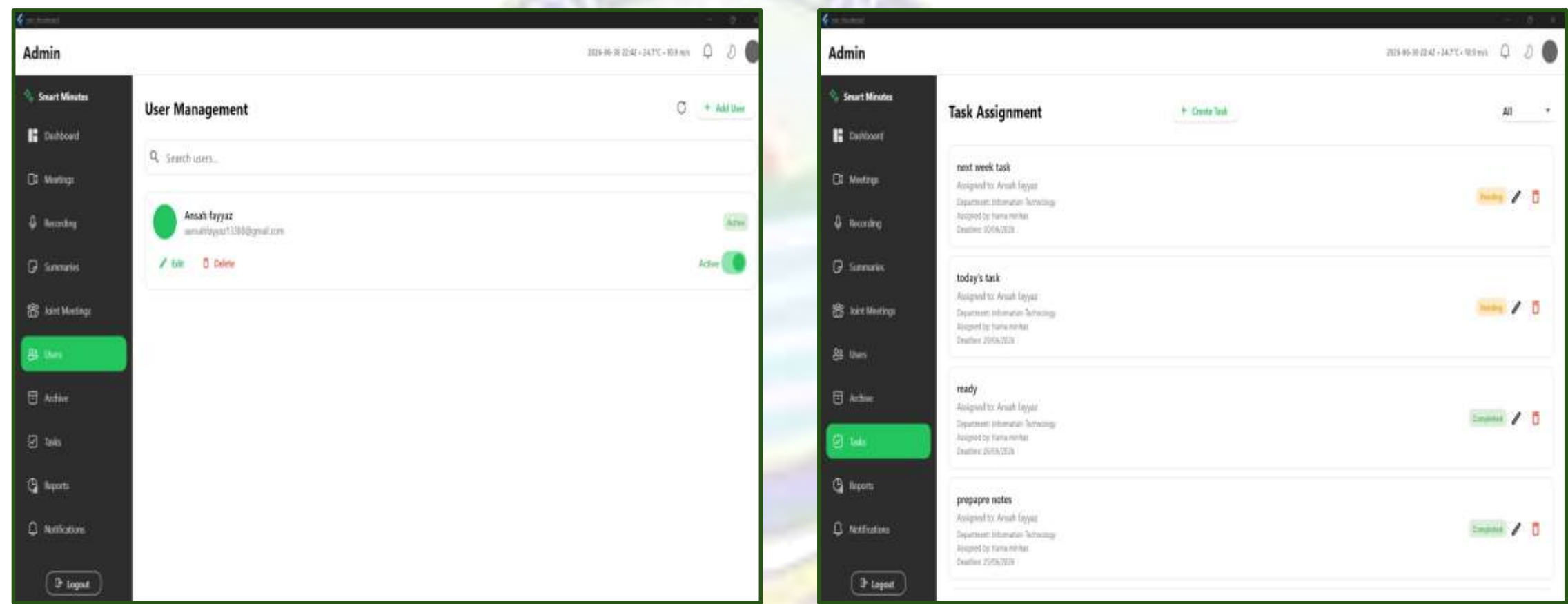
Meeting & recording views



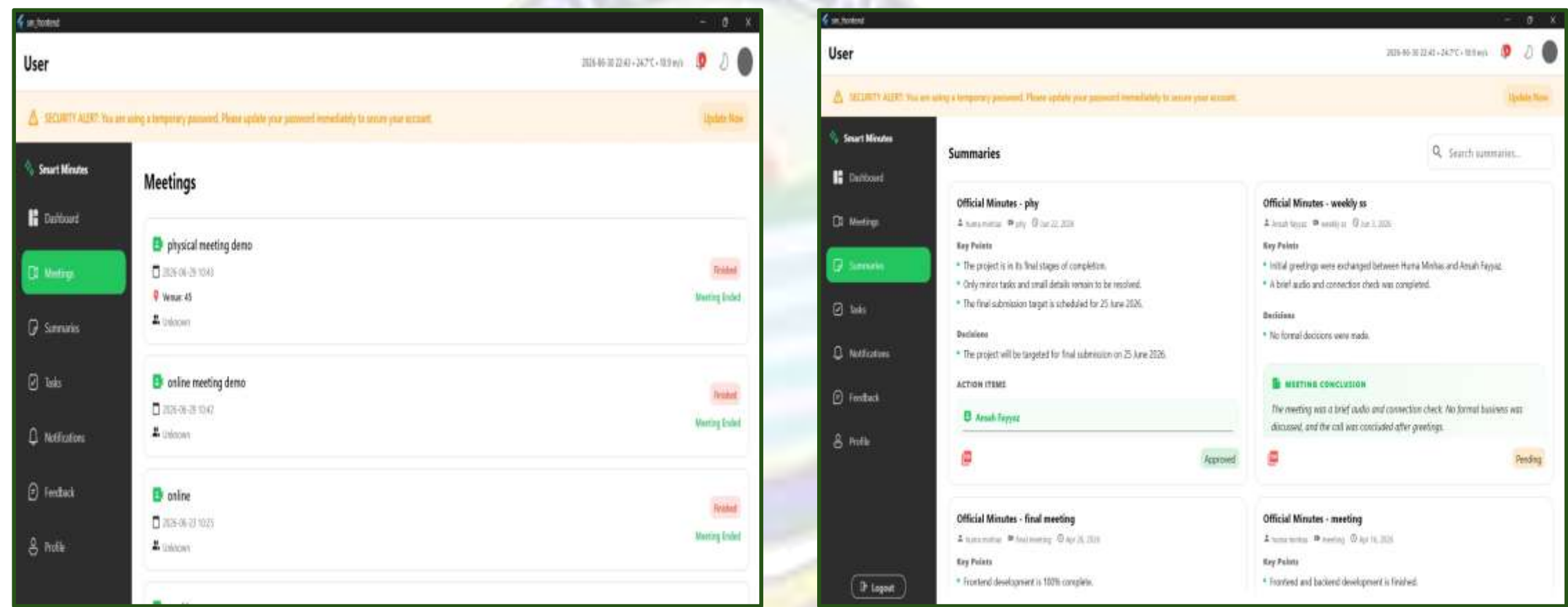
Summary approval workflow



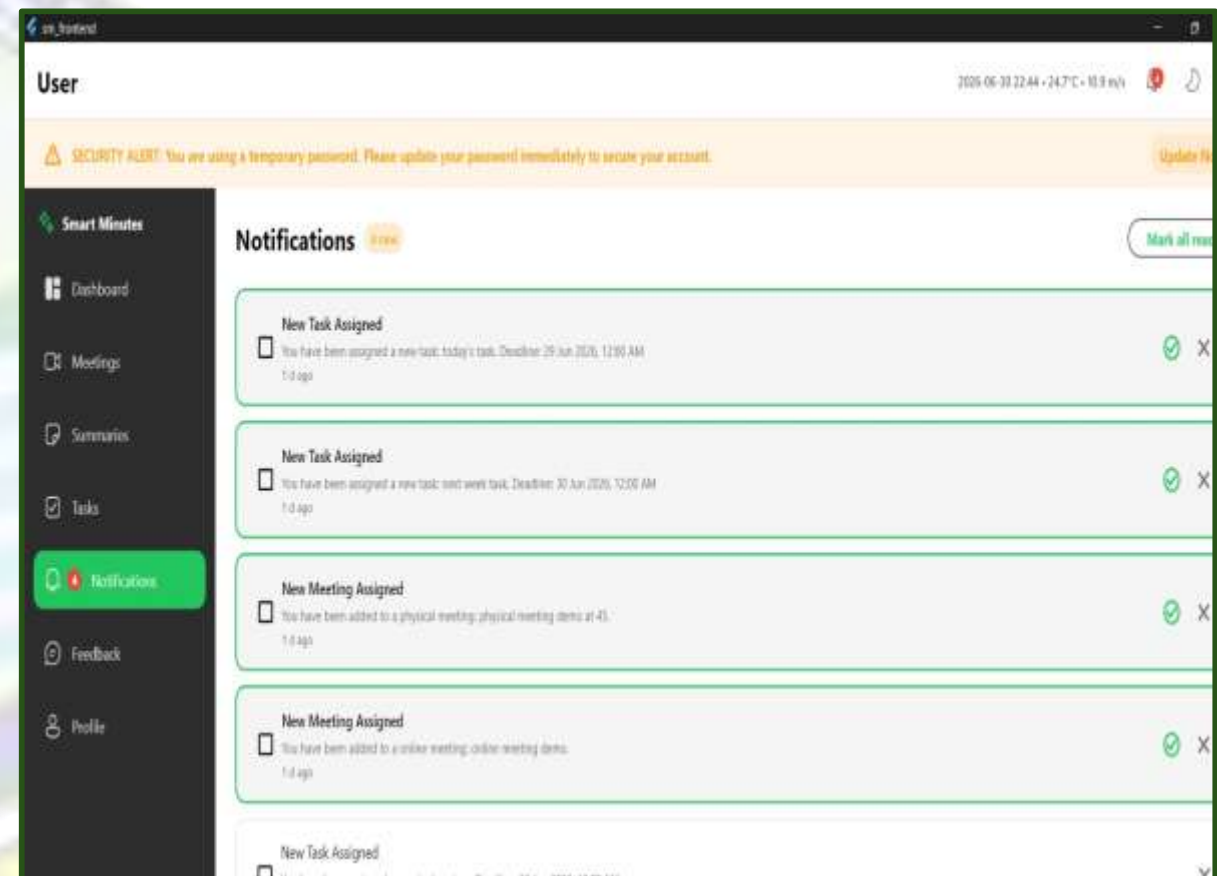
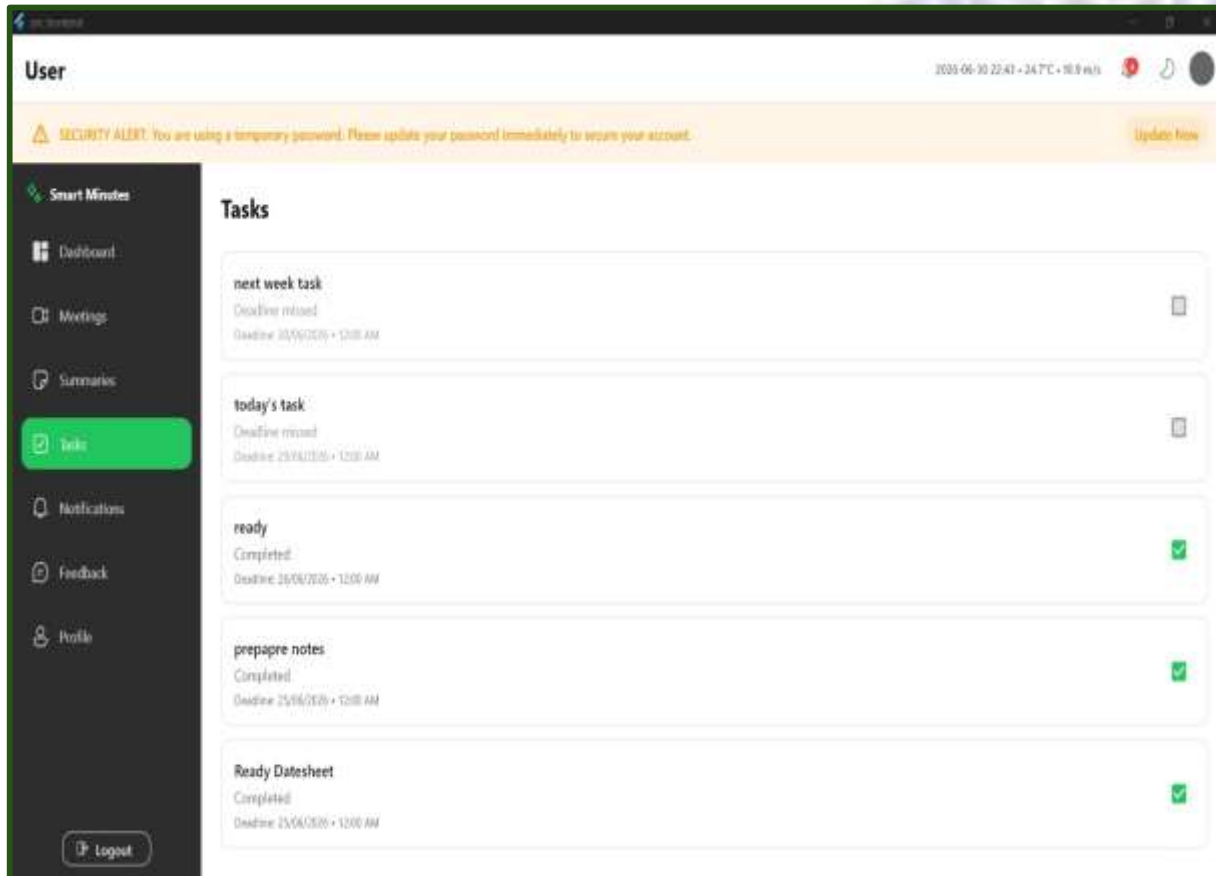
User & task management



User meeting views



Tasks & notifications

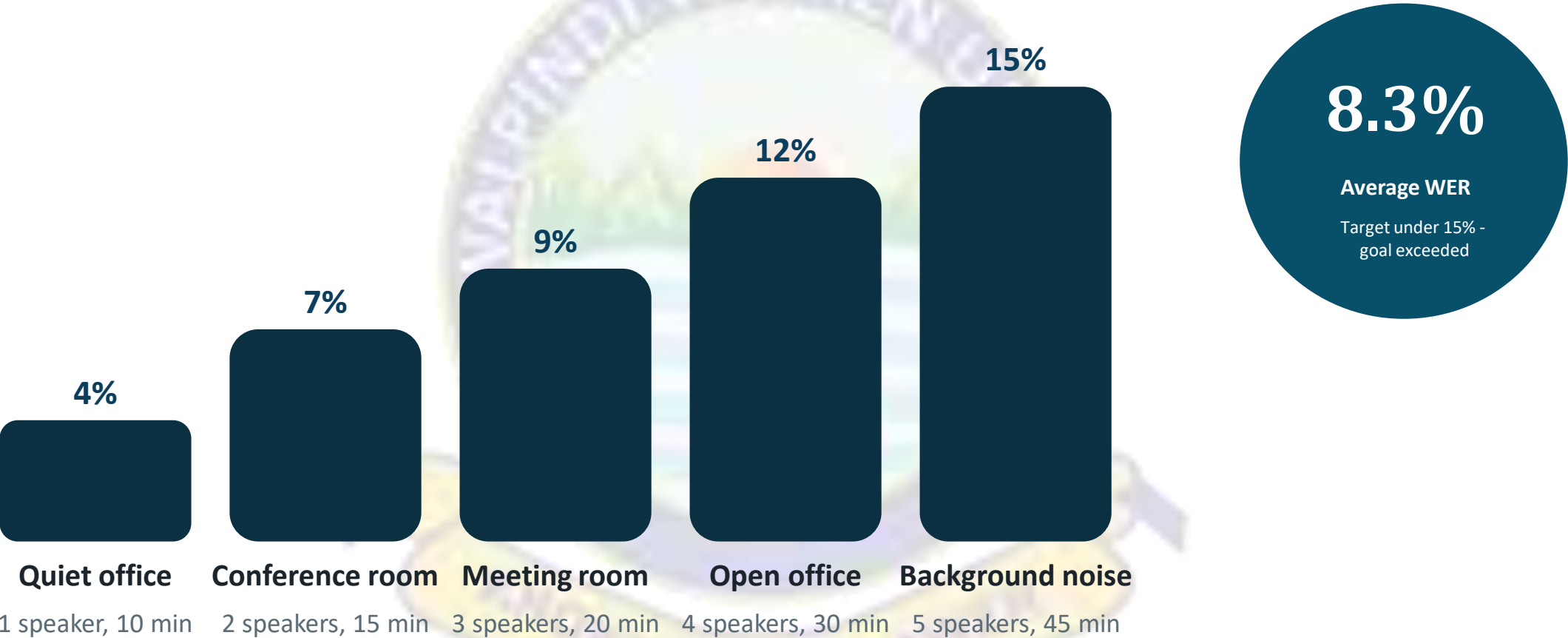


Functional test cases

ID	Module	Test case	Status
TC-01	Auth	Login with valid credentials	Pass
TC-02	Auth	Login with invalid password	Pass
TC-03	Recording	Start meeting recording	Pass
TC-04	Recording	Stop recording and save WAV	Pass
TC-05	STT	Transcribe 10-min audio	Pass
TC-06	AI	Generate meeting summary	Pass
TC-07	Admin	Approve meeting summary	Pass
TC-08	PDF	Export meeting minutes PDF	Pass
TC-09	Email	Send minutes via email	Pass
TC-10	Online	Record online meeting	Pass



Transcription accuracy results



Smart Minutes drives real SDG impact

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Decent work & economic growth

- Automates documentation - frees professionals for high-value productive work
- Reduces meeting overhead costs for SMEs and enterprises
- Women-led tech innovation: 3 jobs created, replicable model for IT Pakistan

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Industry, innovation & infrastructure

- First Pakistani offline-first AI meeting intelligence platform
- Built on a production-grade AI stack: Vosk, Whisper, Gemini 1.5 Pro
- Proves emerging-market universities can build export-ready AI products

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Peace, justice & strong institutions

- Tamper-resistant minutes for government and academic institutions
- Role-based access ensures accountability and transparent decision trails
- Offline-first design enables air-gapped government environments

Smart Minutes is purpose-built for institutions that demand accountability, productivity, and data privacy.

Conclusion & future work

Conclusion

- Complete AI-powered, privacy-centered meeting system
- WER 8.3% - Summary score 4.2/5 - Caption delay 1.4s
- All performance targets met
- Hybrid recording, 3-tier RBAC, automated tasks, and PDF distribution all delivered

Future work

- Speaker diarization using pyannote.audio
- Multi-language support for Vosk and Whisper
- Cloud database migration and mobile app development
- Local LLM integration to reduce Gemini API dependency

The logo of Rawalpindi Women University is a circular emblem. The outer ring contains the text "RAWALPINDI WOMEN UNIVERSITY" in a serif font. The inner circle features a landscape with a sun rising over hills and a body of water. A yellow banner at the bottom of the emblem reads "KNOWLEDGE IS POWER".

THANK YOU

Questions & discussion

Smart Minutes - Ifra Malik, Maha, Zill-e-Huma Minhas

BS Information Technology - Rawalpindi Women University